

# Supplementary Materials: Use of Carabids for the Post-Market Environmental Monitoring of Genetically Modified Crops

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**Table S1.** The sample dates of deployment of pitfall traps at the locality western Slovakia in 2014 and 2015.

| 2014         | 2015         | Stage of maize |
|--------------|--------------|----------------|
| May 15       | May 21       | BBCH 09        |
| June 10      | June 04      | BBCH 11        |
| June 24      | June 18      | BBCH 13        |
| July 8       | July 01      | BBCH 17        |
| July 22      | July 16      | BBCH 34        |
| August 5     | July 30      | BBCH 53        |
| August 19    | August 12    | BBCH 63        |
| September 3  | August 27    | BBCH 69        |
| September 16 | September 10 | BBCH 79        |
| October 1    | September 24 | BBCH 89        |

**Table S2.** The abundance of carabids in the examined localities in 2002–2015.

| Tribe         | Species                                      | Localities         |                    |                    |                    |                     | Total |
|---------------|----------------------------------------------|--------------------|--------------------|--------------------|--------------------|---------------------|-------|
|               |                                              | South<br>Bohemia 1 | South<br>Bohemia 2 | South<br>Bohemia 3 | Central<br>Bohemia | western<br>Slovakia |       |
| Pterostichini | <i>Abax parallelus</i> (Duftschmid, 1812)    | 0                  | 0                  | 0                  | 1                  | 0                   | 1     |
| Harpalini     | <i>Acupalpus meridianus</i> (Linnaeus, 1761) | 0                  | 0                  | 0                  | 1                  | 1                   | 2     |
| Platynini     | <i>Agonum gracilipes</i> (Duftschmid, 1812)  | 0                  | 11                 | 3                  | 0                  | 0                   | 14    |
| Platynini     | <i>Agonum muelleri</i> (Herbst, 1784)        | 42                 | 211                | 3                  | 0                  | 0                   | 256   |
| Platinini     | <i>Agonum sexpunctatum</i> (Linnaeus, 1758)  | 1                  | 5                  | 3                  | 0                  | 0                   | 9     |
| Platynini     | <i>Agonum versutum</i> (Sturm, 1824)         | 0                  | 0                  | 1                  | 0                  | 0                   | 1     |
| Zabrini       | <i>Amara aenea</i> (De Geer, 1774)           | 0                  | 75                 | 6                  | 1                  | 1                   | 83    |
| Zabrini       | <i>Amara apricaria</i> (Paykull, 1790)       | 0                  | 6                  | 0                  | 0                  | 1                   | 7     |
| Zabrini       | <i>Amara aulica</i> (Panzer, 1796)           | 2                  | 2                  | 3                  | 0                  | 0                   | 7     |
| Zabrini       | <i>Amara consularis</i> (Duftschmid 1812)    | 3                  | 1                  | 0                  | 139                | 0                   | 143   |
| Zabrini       | <i>Amara cursitans</i> (Zimmermann, 1831)    | 1                  | 5                  | 0                  | 0                  | 0                   | 6     |
| Zabrini       | <i>Amara eurynota</i> (Panzer, 1797)         | 1                  | 55                 | 0                  | 0                  | 0                   | 56    |
| Zabrini       | <i>Amara familiaris</i> (Duftschmid 1812)    | 0                  | 8                  | 0                  | 0                  | 0                   | 8     |
| Zabrini       | <i>Amara littorea</i> (C.G. Thomson, 1857)   | 1                  | 6                  | 0                  | 0                  | 0                   | 7     |
| Zabrini       | <i>Amara nitida</i> (Sturm, 1825)            | 0                  | 1                  | 0                  | 0                  | 0                   | 1     |
| Zabrini       | <i>Amara ovata</i> (Fabricius, 1792)         | 3                  | 8                  | 0                  | 6                  | 0                   | 17    |
| Zabrini       | <i>Amara similata</i> (Gyllenhal, 1810)      | 0                  | 59                 | 0                  | 0                  | 0                   | 59    |

|            |                                                   |     |     |    |    |      |      |
|------------|---------------------------------------------------|-----|-----|----|----|------|------|
| Zabrini    | <i>Amara tibialis</i> (Paykull, 1798)             | 0   | 2   | 0  | 0  | 0    | 2    |
| Platynini  | <i>Anchomenus dorsalis</i> (Pontoppidan, 1763)    | 3   | 3   | 10 | 12 | 1071 | 1099 |
| Harpalini  | <i>Anisodactylus binotatus</i> (Fabricius, 1787)  | 0   | 0   | 2  | 0  | 3    | 5    |
| Harpalini  | <i>Anisodactylus signatus</i> (Panzer, 1797)      | 0   | 3   | 1  | 2  | 1    | 7    |
| Licinini   | <i>Badister bullatus</i> (Schränk, 1798)          | 0   | 0   | 1  | 3  | 0    | 4    |
| Licinini   | <i>Badister lacertosus</i> (Sturm, 1815)          | 0   | 2   | 0  | 0  | 0    | 2    |
| Licinini   | <i>Badister unipustulatus</i> (Bonelli, 1813)     | 0   | 1   | 0  | 0  | 0    | 1    |
| Bembidiini | <i>Bembidion biguttatum</i> (Fabricius, 1779)     | 0   | 3   | 0  | 0  | 0    | 3    |
| Bembidiini | <i>Bembidion femoratum</i> (Sturm, 1825)          | 20  | 1   | 2  | 0  | 0    | 23   |
| Bembidiini | <i>Bembidion guttula</i> (Fabricius, 1792)        | 0   | 1   | 0  | 0  | 0    | 1    |
| Bembidiini | <i>Bembidion illigeri</i> (Netolitzky, 1914)      | 0   | 2   | 0  | 0  | 0    | 2    |
| Bembidiini | <i>Bembidion lampros</i> (Herbst, 1784)           | 218 | 163 | 18 | 63 | 0    | 462  |
| Bembidiini | <i>Bembidion obtusum</i> (Audinet-Serville, 1821) | 0   | 0   | 0  | 45 | 1    | 46   |
| Bembidiini | <i>Bembidion properans</i> (Stephens, 1828)       | 34  | 34  | 2  | 41 | 3    | 114  |

Table S2. Cont.

| Tribe      | Species                                           | Localities         |                    |                    |                    |                     | Total |
|------------|---------------------------------------------------|--------------------|--------------------|--------------------|--------------------|---------------------|-------|
|            |                                                   | South<br>Bohemia 1 | South<br>Bohemia 2 | South<br>Bohemia 3 | Central<br>Bohemia | western<br>Slovakia |       |
| Bembidiini | <i>Bembidion quadrimaculatum</i> (Linnaeus, 1761) | 813                | 728                | 80                 | 59                 | 0                   | 1680  |
| Brachinini | <i>Brachinus crepitans</i> (Linnaeus, 1758)       | 0                  | 0                  | 0                  | 13                 | 335                 | 348   |
| Brachinini | <i>Brachinus ganglbaueri</i> (Schauberger, 1921)  | 0                  | 0                  | 0                  | 0                  | 294                 | 294   |

|            |                                                     |     |     |     |      |     |      |
|------------|-----------------------------------------------------|-----|-----|-----|------|-----|------|
| Brachinini | <i>Brachinus explodens</i> (Duftschmid, 1812)       | 0   | 0   | 0   | 0    | 71  | 71   |
| Brachinini | <i>Brachinus psophia</i> (Audinet-Serville 1821)    | 0   | 0   | 0   | 0    | 31  | 31   |
| Platynini  | <i>Calathus ambiguus</i> (Paykull, 1790)            | 13  | 3   | 0   | 1    | 14  | 31   |
| Platynini  | <i>Calathus erratus</i> (C. R. Sahlberg, 1827)      | 0   | 6   | 0   | 2    | 0   | 8    |
| Platynini  | <i>Calathus fuscipes</i> (Goeze, 1777)              | 814 | 566 | 473 | 804  | 154 | 2811 |
| Platynini  | <i>Calathus melanocephalus</i> (Linnaeus, 1758)     | 71  | 7   | 13  | 0    | 0   | 91   |
| Carabini   | <i>Calosoma auropunctatum</i> (Herbst, 1784)        | 0   | 0   | 0   | 0    | 2   | 2    |
| Carabini   | <i>Carabus granulatus</i> (Linnaeus, 1758)          | 2   | 493 | 101 | 0    | 0   | 596  |
| Carabini   | <i>Carabus hortensis</i> (Linnaeus, 1758)           | 0   | 2   | 0   | 0    | 0   | 2    |
| Carabini   | <i>Carabus scheidleri</i> (Panzer, 1799)            | 36  | 1   | 0   | 0    | 0   | 37   |
| Carabini   | <i>Carabus violaceus</i> (Linnaeus, 1758)           | 0   | 11  | 2   | 0    | 1   | 14   |
| Callistini | <i>Chlaenius nitidulus</i> (Schrank, 1781)          | 0   | 2   | 0   | 0    | 0   | 2    |
| Clivinini  | <i>Clivina fossor</i> (Linnaeus, 1758)              | 161 | 135 | 29  | 0    | 0   | 325  |
| Platynini  | <i>Dolichus halensis</i> (Schaller, 1783)           | 0   | 0   | 0   | 42   | 52  | 94   |
| Harpalini  | <i>Harpalus affinis</i> (Schrank, 1781)             | 97  | 484 | 117 | 60   | 162 | 920  |
| Harpalini  | <i>Harpalus caspius roubali</i> (Schauberger, 1928) | 0   | 0   | 0   | 0    | 1   | 1    |
| Harpalini  | <i>Harpalus distinguendus</i> (Duftschmid, 1812)    | 0   | 20  | 0   | 124  | 60  | 204  |
| Harpalini  | <i>Harpalus honestus</i> (Duftschmid, 1812)         | 0   | 0   | 0   | 3    | 0   | 3    |
| Harpalini  | <i>Harpalus rubripes</i> (Dufschmid, 1812)          | 0   | 8   | 2   | 2724 | 0   | 2734 |
| Harpalini  | <i>Harpalus signaticornis</i> (Duftschmid in 1812)  | 0   | 1   | 0   | 1    | 0   | 2    |
| Harpalini  | <i>Harpalus smaragdinus</i> (Duftschmid in 1812)    | 1   | 1   | 0   | 0    | 0   | 2    |

|            |                                              |    |    |    |    |    |    |
|------------|----------------------------------------------|----|----|----|----|----|----|
| Harpalini  | <i>Harpalus tardus</i> (Panzer, 1797)        | 0  | 0  | 0  | 0  | 1  | 1  |
| Platynini  | <i>Laemostenus terricola</i> (Herbst, 1784)  | 0  | 0  | 0  | 0  | 38 | 38 |
| Trechini   | <i>Lasiotrechus discus</i> (Fabricius, 1792) | 1  | 0  | 0  | 0  | 0  | 1  |
| Nebrini    | <i>Leistus ferrugineus</i> (Linnaeus, 1758)  | 0  | 0  | 0  | 0  | 15 | 15 |
| Loricerini | <i>Loricera pilicornis</i> (Fabricius, 1775) | 37 | 2  | 11 | 1  | 0  | 51 |
| Lebiini    | <i>Microlestes maurus</i> (Sturm, 1827)      | 4  | 16 | 1  | 12 | 4  | 37 |
| Lebiini    | <i>Microlestes minutulus</i> (Goeze 1777)    | 3  | 8  | 0  | 0  | 0  | 11 |

Table S2. Cont.

| Tribe         | Species                                               | Localities         |                    |                    |                    |                     | Total  |
|---------------|-------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|---------------------|--------|
|               |                                                       | South<br>Bohemia 1 | South<br>Bohemia 2 | South<br>Bohemia 3 | Central<br>Bohemia | western<br>Slovakia |        |
| Nebtriini     | <i>Nebria brevicollis</i> (Fabricius, 1792)           | 19                 | 0                  | 1                  | 0                  | 0                   | 20     |
| Notiophilini  | <i>Notiophilus aquaticus</i> (Linnaeus, 1758)         | 0                  | 1                  | 1                  | 0                  | 0                   | 2      |
| Notiophilini  | <i>Notiophilus palustris</i> (Duftschmid, 1812)       | 3                  | 0                  | 0                  | 59                 | 1                   | 63     |
| Notiophilini  | <i>Notiophilus pusillus</i> (G.R.Waterhouse, 1833)    | 0                  | 0                  | 0                  | 13                 | 0                   | 13     |
| Harpalini     | <i>Ophonus azureus</i> (Fabricius, 1775)              | 0                  | 2                  | 1                  | 0                  | 0                   | 3      |
| Harpalini     | <i>Ophonus nitidulus</i> (Stephens, 1828)             | 0                  | 0                  | 0                  | 1                  | 0                   | 1      |
| Platynini     | <i>Platynus assimilis</i> (Paykull, 1790)             | 0                  | 2                  | 1                  | 0                  | 0                   | 3      |
| Pterostichini | <i>Poecilus cupreus</i> (Linnaeus, 1758)              | 874                | 13 234             | 1467               | 250                | 150                 | 15 975 |
| Pterostichini | <i>Poecilus lepidus</i> (Leske 1785)                  | 0                  | 0                  | 0                  | 0                  | 63                  | 63     |
| Pterostichini | <i>Poecilus sericeus</i> (Fischer von Waldheim, 1824) | 0                  | 0                  | 0                  | 0                  | 77                  | 77     |

|               |                                                     |      |      |        |     |      |        |
|---------------|-----------------------------------------------------|------|------|--------|-----|------|--------|
| Pterostichini | <i>Poecilus versicolor</i> (Sturm, 1824)            | 1    | 1499 | 207    | 3   | 0    | 1710   |
| Pterostichini | <i>Pseudoophonus griseus</i> (Panzer, 1796)         | 0    | 17   | 0      | 5   | 0    | 22     |
| Pterostichini | <i>Pseudoophonus rufipes</i> (De Geer)              | 239  | 932  | 144    | 0   | 6556 | 7871   |
| Pterostichini | <i>Pterostichus melanarius</i> (Illiger, 1798)      | 1740 | 2978 | 12 506 | 995 | 78   | 18 297 |
| Pterostichini | <i>Pterostichus melas</i> (Creutzer, 1799)          | 0    | 0    | 0      | 0   | 1    | 1      |
| Pterostichini | <i>Pterostichus niger</i> (Schaller, 1783)          | 2    | 87   | 21     | 0   | 1    | 111    |
| Pterostichini | <i>Pterostichus nigrata</i> (Paykull, 1790)         | 0    | 52   | 11     | 0   | 0    | 63     |
| Pterostichini | <i>Pterostichus ovoideus</i> (Sturm, 1824)          | 0    | 2    | 0      | 0   | 0    | 2      |
| Pterostichini | <i>Pterostichus strenuus</i> (Panzer, 1796)         | 13   | 10   | 0      | 0   | 0    | 23     |
| Pterostichini | <i>Pterostichus vernalis</i> (Panzer, 1795)         | 7    | 3    | 0      | 1   | 0    | 11     |
| Pterostichini | <i>Stomis pumicatus</i> (Panzer, 1795)              | 0    | 0    | 0      | 4   | 0    | 4      |
| Trechini      | <i>Trechus quadristriatus</i> (Schränk, 1781)       | 204  | 33   | 329    | 160 | 115  | 841    |
| Harpalini     | <i>Trichotichnus laevicollis</i> (Duftschmid, 1812) | 0    | 1    | 0      | 0   | 0    | 1      |
| Zabrinini     | <i>Zabrus tenebrioides</i> (Goeze, 1777)            | 0    | 0    | 0      | 180 | 42   | 222    |

**Table S3.** Incidence and functional traits of carabids<sup>1</sup> captured in localities South Bohemia 1, 2 and 3, Central Bohemia and western Slovakia in 2002–2015.

| Tribe         | Species                                         | Incidence     | Body size | Habitat affinity | Humidity affinity | Breeding period | Food specialization |
|---------------|-------------------------------------------------|---------------|-----------|------------------|-------------------|-----------------|---------------------|
| Pterostichini | <i>Abax parallelus</i><br>(Duftschmid, 1812)    | Least concern | B         | Silvicolous      | Hygrophilous      | Spring          | Carnivorous         |
| Harpalini     | <i>Acupalpus meridianus</i><br>(Linnaeus, 1761) | Least concern | D         | Open biotopes    | Mesophilous       | Spring          | Omnivorous          |
| Platynini     | <i>Agonum gracilipes</i><br>(Duftschmid, 1812)  | Least concern | C         | Silvicolous      | Eurytopic         | Spring          | Carnivorous         |
| Platynini     | <i>Agonum muelleri</i><br>(Herbst, 1784)        | Least concern | C         | Eurytopic        | Hygrophilous      | Spring          | Carnivorous         |
| Platinini     | <i>Agonum sexpunctatum</i><br>(Linnaeus, 1758)  | Least concern | C         | Eurytopic        | Hygrophilous      | Spring          | Carnivorous         |
| Platynini     | <i>Agonum versutum</i><br>(Sturm, 1824)         | Least concern | C         | Silvicolous      | Hygrophilous      | Spring          | Carnivorous         |
| Zabrini       | <i>Amara aenea</i><br>(De Geer, 1774)           | Least concern | C         | Open biotopes    | Xerophilous       | Spring          | Omnivorous          |
| Zabrini       | <i>Amara apricaria</i><br>(Paykull, 1790)       | Least concern | C         | Open biotopes    | Xerophilous       | Autumn          | Omnivorous          |
| Zabrini       | <i>Amara aulica</i><br>(Panzer, 1796)           | Least concern | B         | Open biotopes    | Mesophilous       | Autumn          | Omnivorous          |
| Zabrini       | <i>Amara consularis</i><br>(Duftschmid 1812)    | Least concern | C         | Eurytopic        | Eurytopic         | Autumn          | Omnivorous          |
| Zabrini       | <i>Amara cursitans</i><br>(Zimmermann, 1831)    | Least concern | C         | Eurytopic        | Eurytopic         | Autumn          | Omnivorous          |

|         |                                               |               |   |               |              |        |            |
|---------|-----------------------------------------------|---------------|---|---------------|--------------|--------|------------|
| Zabrini | <i>Amara eurynota</i><br>(Panzer, 1797)       | Least concern | C | Open biotopes | Mesophilous  | Spring | Omnivorous |
| Zabrini | <i>Amara familiaris</i><br>(Duftschmid 1812)  | Least concern | C | Open biotopes | Mesophilous  | Spring | Omnivorous |
| Zabrini | <i>Amara littorea</i><br>(C.G. Thomson, 1857) | Least concern | C | Open biotopes | Xerophilous  | Spring | Omnivorous |
| Zabrini | <i>Amara nitida</i><br>(Sturm, 1825)          | Least concern | C | Open biotopes | Xerophilous  | Spring | Omnivorous |
| Zabrini | <i>Amara ovata</i><br>(Fabricius, 1792)       | Least concern | C | Silvicolous   | Hygrophilous | Spring | Omnivorous |

**Table S3.** *Cont.*

| Tribe     | Species                                             | Incidence     | Body size | Habitat affinity | Humidity affinity | Breeding period | Food specialization |
|-----------|-----------------------------------------------------|---------------|-----------|------------------|-------------------|-----------------|---------------------|
| Zabrini   | <i>Amara similata</i><br>(Gyllenhal, 1810)          | Least concern | C         | Open biotopes    | Mesophilous       | Spring          | Omnivorous          |
| Zabrini   | <i>Amara tibialis</i><br>(Paykull, 1798)            | Least concern | D         | Open biotopes    | Mesophilous       | Spring/summer   | Omnivorous          |
| Platynini | <i>Anchomenus dorsalis</i><br>(Pontoppidan, 1763)   | Least concern | C         | Open biotopes    | Hygrophilous      | Spring          | Carnivorous         |
| Harpalini | <i>Anisodactylus binotatus</i><br>(Fabricius, 1787) | Least concern | C         | Open biotopes    | Mesophilous       | Spring          | Omnivorous          |
| Harpalini | <i>Anisodactylus signatus</i><br>(Panzer, 1797)     | Least concern | B         | Open biotopes    | Hygrophilous      | Spring          | Omnivorous          |
| Licinini  | <i>Badister bullatus</i>                            | Least concern | D         | Eurytopic        | Hygrophilous      | Spring          | Carnivorous         |

|                 |                                                          |               |   |               |              |                        |             |
|-----------------|----------------------------------------------------------|---------------|---|---------------|--------------|------------------------|-------------|
| (Schränk, 1798) |                                                          |               |   |               |              |                        |             |
| Licinini        | <i>Badister lacertosus</i><br>(Sturm, 1815)              | Least concern | C | Silvicolous   | Hygrophilous | Spring                 | Carnivorous |
| Licinini        | <i>Badister unipustulatus</i><br>(Bonelli, 1813)         | Least concern | C | Silvicolous   | Hygrophilous | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion biguttatum</i><br>(Fabricius, 1779)         | Least concern | D | Silvicolous   | Hygrophilous | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion femoratum</i><br>(Sturm, 1825)              | Least concern | D | Eurytopic     | Mesophilous  | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion guttula</i><br>(Fabricius, 1792)            | Least concern | D | Eurytopic     | Hygrophilous | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion illigeri</i><br>(Netolitzky, 1914)          | Least concern | D | Eurytopic     | Hygrophilous | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion lampros</i><br>(Herbst, 1784)               | Least concern | D | Open biotopes | Eurytopic    | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion obtusum</i><br>(Audinet-Serville, 1821)     | Least concern | D | Eurytopic     | Eurytopic    | Autumn/early<br>spring | Carnivorous |
| Bembidiini      | <i>Bembidion properans</i><br>(Stephens, 1828)           | Least concern | D | Open biotopes | Hygrophilous | Spring                 | Carnivorous |
| Bembidiini      | <i>Bembidion<br/>quadrimaculatum</i><br>(Linnaeus, 1761) | Least concern | D | Open biotopes | Eurytopic    | Spring                 | Carnivorous |

**Table S3.** *Cont.*

| Tribe | Species | Incidence | Body size | Habitat affinity | Humidity affinity | Breeding period | Food<br>specialization |
|-------|---------|-----------|-----------|------------------|-------------------|-----------------|------------------------|
|-------|---------|-----------|-----------|------------------|-------------------|-----------------|------------------------|

|            |                                                     |                                     |   |               |              |        |             |
|------------|-----------------------------------------------------|-------------------------------------|---|---------------|--------------|--------|-------------|
| Brachinini | <i>Brachinus crepitans</i><br>(Linnaeus, 1758)      | Least concern                       | C | Open biotopes | Xerophilous  | Summer | Carnivorous |
| Brachinini | <i>Brachinus ganglbaueri</i><br>(Schauberger, 1921) | Least concern                       | C | Open biotopes | Hygrophilous | Spring | Carnivorous |
| Brachinini | <i>Brachinus explodens</i><br>(Duftschmid, 1812)    | Least concern                       | D | Open biotopes | Xerophilous  | Spring | Carnivorous |
| Brachinini | <i>Brachinus psophia</i><br>(Audinet-Serville 1821) | CZ: endangered<br>SK: least concern | C | Open biotopes | Hygrophilous | Spring | Carnivorous |
| Platynini  | <i>Calathus ambiguus</i><br>(Paykull, 1790)         | Least concern                       | C | Open biotopes | Xerophilous  | Autumn | Carnivorous |
| Platynini  | <i>Calathus erratus</i><br>(C. R. Sahlberg, 1827)   | Least concern                       | C | Open biotopes | Xerophilous  | Autumn | Carnivorous |
| Platynini  | <i>Calathus fuscipes</i><br>(Goeze, 1777)           | Least concern                       | B | Open biotopes | Xerophilous  | Autumn | Carnivorous |
| Platynini  | <i>Calathus melanocephalus</i><br>(Linnaeus, 1758)  | Least concern                       | C | Open biotopes | Mesophilous  | Autumn | Carnivorous |
| Carabini   | <i>Calosoma auropunctatum</i><br>(Herbst, 1784)     | CZ: vulnerable<br>SK: endangered    | A | Open biotopes | Xerophilous  | Spring | Carnivorous |
| Carabini   | <i>Carabus granulatus</i><br>(Linnaeus, 1758)       | Least concern                       | B | Silvicolous   | Hygrophilous | Spring | Carnivorous |
| Carabini   | <i>Carabus hortensis</i><br>(Linnaeus, 1758)        | Least concern                       | A | Silvicolous   | Hygrophilous | Autumn | Carnivorous |
| Carabini   | <i>Carabus scheidleri</i><br>(Panzer, 1799)         | Least concern                       | A | Silvicolous   | Eurytopic    | Autumn | Carnivorous |
| Carabini   | <i>Carabus violaceus</i><br>(Linnaeus, 1758)        | Least concern                       | A | Silvicolous   | Eurytopic    | Autumn | Carnivorous |

|            |                                               |               |   |               |              |                        |             |
|------------|-----------------------------------------------|---------------|---|---------------|--------------|------------------------|-------------|
| Callistini | <i>Chlaenius nitidulus</i><br>(Schränk, 1781) | Least concern | B | Open biotopes | Hygrophilous | Spring                 | Carnivorous |
| Clivinini  | <i>Clivina fossor</i><br>(Linnaeus, 1758)     | Least concern | C | Open biotopes | Hygrophilous | Spring                 | Carnivorous |
| Platynini  | <i>Dolichus halensis</i><br>(Schaller, 1783)  | Least concern | B | Open biotopes | Xerophilous  | Late summer/<br>autumn | Carnivorous |

**Table S3.** *Cont.*

| Tribe     | Species                                                | Incidence     | Body size | Habitat affinity | Humidity affinity | Breeding period          | Food specialization |
|-----------|--------------------------------------------------------|---------------|-----------|------------------|-------------------|--------------------------|---------------------|
| Harpalini | <i>Harpalus affinis</i><br>(Schränk, 1781)             | Least concern | C         | Open biotopes    | Eurytopic         | Spring/summer/<br>autumn | Omnivorous          |
| Harpalini | <i>Harpalus caspius roubali</i><br>(Schauberger, 1928) | Least concern | B         | Open biotopes    | Xerophilous       | Autumn                   | Omnivorous          |
| Harpalini | <i>Harpalus distinguendus</i><br>(Duftschmid, 1812)    | Least concern | C         | Open biotopes    | Eurytopic         | Spring/summer            | Omnivorous          |
| Harpalini | <i>Harpalus honestus</i><br>(Duftschmid, 1812)         | Least concern | C         | Open biotopes    | Xerophilous       | Spring                   | Omnivorous          |
| Harpalini | <i>Harpalus rubripes</i><br>(Duftschmid, 1812)         | Least concern | C         | Open biotopes    | Eurytopic         | Spring                   | Omnivorous          |
| Harpalini | <i>Harpalus signaticornis</i><br>(Duftschmid in 1812)  | Least concern | C         | Open biotopes    | Xerophilous       | Spring/autumn            | Omnivorous          |
| Harpalini | <i>Harpalus smaragdinus</i><br>(Duftschmid in 1812)    | Least concern | C         | Open biotopes    | Xerophilous       | Spring/autumn            | Omnivorous          |
| Harpalini | <i>Harpalus tardus</i>                                 | Least concern | C         | Open biotopes    | Eurytopic         | Spring                   | Omnivorous          |

|                |                                                  |                                          |   |               |              |               |             |
|----------------|--------------------------------------------------|------------------------------------------|---|---------------|--------------|---------------|-------------|
| (Panzer, 1797) |                                                  |                                          |   |               |              |               |             |
| Platynini      | <i>Laemostenus terricola</i><br>(Herbst, 1784)   | CZ: near threatened<br>SK: least concern | B | Open biotopes | Xerophilous  | Autumn        | Carnivorous |
| Trechini       | <i>Lasiotrechus discus</i><br>(Fabricius, 1792)  | Least concern                            | D | Eurytopic     | Hygrophilous | Spring        | Carnivorous |
| Nebrini        | <i>Leistus ferrugineus</i><br>(Linnaeus, 1758)   | Least concern                            | C | Eurytopic     | Hygrophilous | Autumn        | Carnivorous |
| Loricerini     | <i>Loricera pilicornis</i><br>(Fabricius, 1775)  | Least concern                            | C | Eurytopic     | Hygrophilous | Spring/summer | Carnivorous |
| Lebiini        | <i>Microlestes maurus</i><br>(Sturm, 1827)       | Least concern                            | D | Eurytopic     | Eurytopic    | Spring        | Omnivorous  |
| Lebiini        | <i>Microlestes minutulus</i><br>(Goeze 1777)     | Least concern                            | D | Eurytopic     | Eurytopic    | Spring        | Carnivorous |
| Nebtriini      | <i>Nebria brevicollis</i><br>(Fabricius, 1792)   | Least concern                            | B | Eurytopic     | Hygrophilous | Autumn        | Carnivorous |
| Notiophilini   | <i>Notiophilus aquaticus</i><br>(Linnaeus, 1758) | Least concern                            | D | Silvicolous   | Hygrophilous | Spring/summer | Carnivorous |

Table S3. Cont.

| Tribe        | Species                                               | Incidence     | Body size | Habitat affinity | Humidity affinity | Breeding period | Food specialization |
|--------------|-------------------------------------------------------|---------------|-----------|------------------|-------------------|-----------------|---------------------|
| Notiophilini | <i>Notiophilus palustris</i><br>(Duftschmid, 1812)    | Least concern | D         | Silvicolous      | Hygrophilous      | Spring          | Carnivorous         |
| Notiophilini | <i>Notiophilus pusillus</i><br>(G.R.Waterhouse, 1833) | Least concern | D         | Eurytopic        | Eurytopic         | Spring          | Carnivorous         |
| Harpalini    | <i>Ophonus azureus</i>                                | Least concern | C         | Open biotopes    | Xerophilous       | Autumn          | Granivorous         |

|               |                                                          |                                     |   |               |              |        |             |
|---------------|----------------------------------------------------------|-------------------------------------|---|---------------|--------------|--------|-------------|
|               | (Fabricius, 1775)                                        |                                     |   |               |              |        |             |
| Harpalini     | <i>Ophonus nitidulus</i><br>(Stephens, 1828)             | Least concern                       | C | Open biotopes | Xerophilous  | Spring | Granivorous |
| Platynini     | <i>Platynus assimilis</i><br>(Paykull, 1790)             | Least concern                       | C | Silvicolous   | Hygrophilous | Spring | Carnivorous |
| Pterostichini | <i>Poecilus cupreus</i><br>(Linnaeus, 1758)              | Least concern                       | B | Eurytopic     | Eurytopic    | Spring | Omnivorous  |
| Pterostichini | <i>Poecilus lepidus</i><br>(Leske 1785)                  | Least concern                       | B | Eurytopic     | Xerophilous  | Spring | Carnivorous |
| Pterostichini | <i>Poecilus sericeus</i><br>(Fischer von Waldheim, 1824) | CZ: vulnerable SK:<br>least concern | B | Eurytopic     | Mesophilous  | Spring | Carnivorous |
| Pterostichini | <i>Poecilus versicolor</i><br>(Sturm, 1824)              | Least concern                       | C | Open biotopes | Hygrophilous | Spring | Carnivorous |
| Pterostichini | <i>Pseudoophonus griseus</i><br>(Panzer, 1796)           | Least concern                       | C | Open biotopes | Mesophilous  | Autumn | Omnivorous  |
| Pterostichini | <i>Pseudoophonus rufipes</i><br>(De Geer)                | Least concern                       | B | Open biotopes | Eurytopic    | Autumn | Omnivorous  |
| Pterostichini | <i>Pterostichus melanarius</i><br>(Illiger, 1798)        | Least concern                       | B | Eurytopic     | Mesophilous  | Autumn | Carnivorous |
| Pterostichini | <i>Pterostichus melas</i><br>(Creutzer, 1799)            | Least concern                       | B | Silvicolous   | Mesophilous  | Autumn | Carnivorous |
| Pterostichini | <i>Pterostichus niger</i><br>(Schaller, 1783)            | Least concern                       | B | Silvicolous   | Hygrophilous | Autumn | Carnivorous |
| Pterostichini | <i>Pterostichus nigrita</i><br>(Paykull, 1790)           | Least concern                       | B | Eurytopic     | Hygrophilous | Spring | Carnivorous |
| Pterostichini | <i>Pterostichus ovoideus</i>                             | Least concern                       | C | Silvicolous   | Hygrophilous | Spring | Carnivorous |

(Sturm, 1824)

Table S3. Cont.

| Tribe         | Species                                                | Incidence     | Body size | Habitat affinity | Humidity affinity | Breeding period | Food specialization |
|---------------|--------------------------------------------------------|---------------|-----------|------------------|-------------------|-----------------|---------------------|
| Pterostichini | <i>Pterostichus strenuus</i><br>(Panzer, 1796)         | Least concern | C         | Silvicolous      | Hygrophilous      | Spring          | Carnivorous         |
| Pterostichini | <i>Pterostichus vernalis</i><br>(Panzer, 1795)         | Least concern | C         | Eurytopic        | Hygrophilous      | Spring/summer   | Carnivorous         |
| Pterostichini | <i>Stomis pumicatus</i><br>(Panzer, 1795)              | Least concern | C         | Eurytopic        | Hygrophilous      | Spring          | Carnivorous         |
| Trechini      | <i>Trechus quadristriatus</i><br>(Schränk, 1781)       | Least concern | D         | Open biotopes    | Mesophilous       | Autumn          | Carnivorous         |
| Harpalini     | <i>Trichotichnus laevicollis</i><br>(Duftschmid, 1812) | Least concern | C         | Silvicolous      | Mesophilous       | Spring/autumn   | Omnivorous          |
| Zabrinini     | <i>Zabrus tenebrioides</i><br>(Goeze, 1777)            | Least concern | B         | Open biotopes    | Xerophilous       | Autumn          | Omnivorous          |

<sup>1</sup> The tribe, species, body size, habitat and humidity affinity, breeding period [1], incidence [2,3] and food specializations [4] were determined. Body size (mid-range): A: >22 mm, B: 11–21.9 mm, C: 6–10.9 mm, D: <5.9 mm; Silvicolous: preferring woodlands; Open biotopes: preferring open areas; Eurytopic: adaptable to various environmental conditions; Hygrophilous: preferring moist places; Mesophilous: preferring intermediate or moderate environmental conditions, avoid extremes of moisture or dryness; Xerophilous: preferring dry environmental conditions.

## References

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